

Solution Sessional I exam

Question 1. A beverage company makes three types of drinks — A (x_1 litres), B (x_2 litres), and C (x_3 litres) that are blended into a single limited-edition drink. The number of units of drinks contributed to recipe balance per liter is 1 unit of A, 2 units of B and 1 unit of C and the mixture must total 6 units. The marketing team insists the batch must score at least 25 quality points. A gives 3 points per liter, B gives 5, C gives 4 points per liter. The cost per unit of A, B and C is \$2, \$3 and \$2 respectively. Find the optimal mix of the drinks A, B and C.

Solution:

$$\text{Minimize } Z = 2x_1 + 3x_2 + 2x_3$$

$$x_1 + 2x_2 + x_3 = 6$$

$$3x_1 + 5x_2 + 4x_3 \geq 25$$

$$x_1, x_2, x_3 \geq 0.$$

One possible way to solve this problem is two phase method.

Initial Table - Phase I

Table 1	C_j	0	0	0	0	1	1	
C_b	Base	X_1	X_2	X_3	S_1	A_1	A_2	R
1	A_1	1	2	1	0	1	0	6
1	A_2	3	5	4	-1	0	1	25
	Z	4	7	5	-1	0	0	31

Table 2	C_j	0	0	0	0	1	1	
C_b	Base	X_1	X_2	X_3	S_1	A_1	A_2	R
0	X_2	1/2	1	1/2	0	1/2	0	3
1	A_2	1/2	0	3/2	-1	-5/2	1	10
	Z	1/2	0	3/2	-1	-7/2	0	10

Table 3	C_j	0	0	0	0	1	1	
C_b	Base	X_1	X_2	X_3	S_1	A_1	A_2	R
0	X_3	1	2	1	0	1	0	6
1	A_2	-1	-3	0	-1	-4	1	1
	Z	-1	-3	0	-1	-5	0	1

The iterations of the first phase have been completed and there are artificial variables in the base with values strictly greater than 0, so the problem has no solution. (4+5+1)

Question 2: For the linear model given below, enumerate all solutions. Classify them as basic, non-basic, feasible and infeasible. Verify your findings by showing these solutions on the graphical solution space.

$$\text{Max } Z = 5x - 10y, \quad 2x + 5y \leq 80, \quad x + y \leq 20 \quad x, y \geq 0.$$

Solution: Convert to standard form by adding slack variables

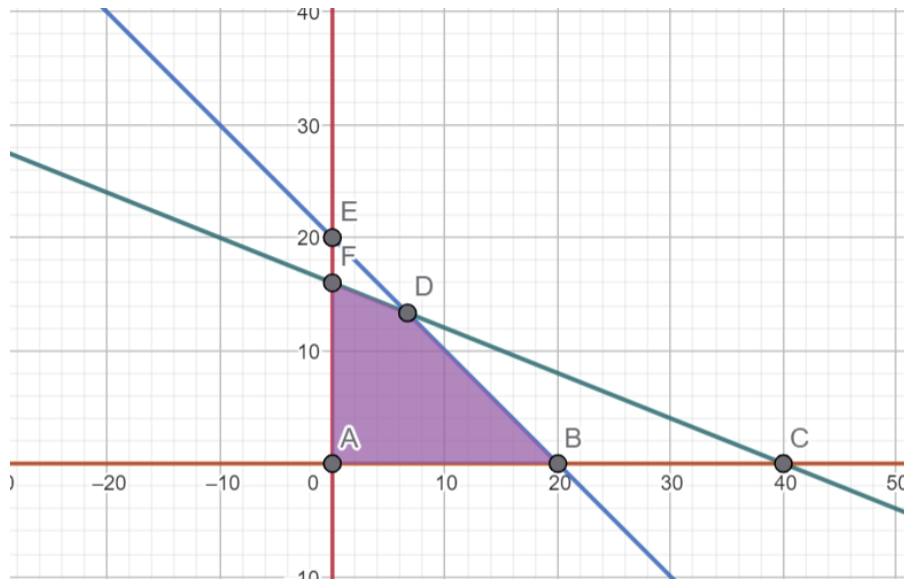
$$2x + 5y + s_1 = 80$$

$$x + y + s_2 = 20.$$

All basic solutions, optimal corner and objective values at feasible corners:

Basis	x	y	s_1	s_2	z	Classification
$\{x, y\}$	20/3	40/3	0	0	-100	feasible
$\{x, s_1\}$	20	0	40	0	100	Feasible and optimal
$\{x, s_2\}$	40	0	0	-20	--	infeasible
$\{y, s_1\}$	0	20	-20	0	--	infeasible
$\{y, s_2\}$	0	16	0	4	-160	feasible
$\{s_1, s_2\}$	0	0	80	20	0	feasible

(5+5 marks)



Question 3: A basic feasible and optimal solution for the linear model

$$\max z = 3x + 2y$$

$$x + y \leq 10$$

$$2x + y \leq 14$$

$$x, y \geq 0.$$

is $x = 4, y = 6$.

Find

- The range so that the current solution remains optimal. If the objective function is replaced by $\max z = 10x + 3y$, will the current solution remain optimal?
- Find the dual price of both constraints. If you are suggested to decrease the capacity of one constraint, what will be your choice?

Solution: For $x = 4, y = 6, z = 24$.

- Consider a general objective function

$$\max z = c_1x_1 + c_2x_2$$

The optimality range is determined by $\frac{c_1}{c_2}$

For constraint 1, $\frac{c_1}{c_2} = 1$

and for constraint 2, $\frac{c_1}{c_2} = 2$.

Thus $1 \leq \frac{c_1}{c_2} \leq 2$. (2 marks)

If $\max z = 10x + 3y, \frac{c_1}{c_2} = \frac{10}{3} = 3.33$ that lies outside the optimal range so the current solution will not remain optimal. (2 marks)

- The dual price can be computed by solving the equations

$$u + 2v = 3$$

$$u + v = 2$$

that yields $u = v = 1$

or alternatively, increasing RHS of one constraint, keep the other unchanged and solving the two equations simultaneously.

Since the dual price of both constraints is same, anyone can be chosen. (2.5+2.5+1)